MILITARY SPECIFICATION

INDICATOR, MACH NUMBER, SENSITIVE, SERVOED, RANGE .5 TO 1.0, TYPE ME-5

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of servo operated, sensitive Mach number indicator, designated Type ME-5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method Of
DOD-D-1000	Drawing, Engineering And Associated List
MIL-T-5350	Transmitters, Synchro, Operated, Aircraft, General Specification For
MIL-E-5400	Electronic Equipment, Aerospace, General Specification For
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads; Standard, Optimum Selection Series. General Specification For
MIL-L-25142	Luminescent Material, Fluorescent

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595 Color

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810C	Environmental Test Methods And Engineering Guidelines
MIL-STD-838	Precedence For The Selection Of Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-2073-1	DOD Material Procedures For Development and Application Of Packaging Requirements
MS28105	Cover Glass, Aircraft Instrument Dial
MS33549	Case, Instrument, 2-3/4 Dial, With Sump, Standard Dimensions For
MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing and Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.6 and 6.3).

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3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial parts, having suitable properties may be used provided they can be replaced by the standard parts (MS or AN) without alteration and the corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the contracting activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt fog or atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the instrument, except where magnetic materials are essential.

3.3.4 Non-ferrous materials. Non-ferrous materials shall be used in fabrication of all parts of the indicator except where use of ferrous materials is essential.

3.3.5 Toxic and corrosive fumes. The materials as installed in the indicator and under the test conditions specified in section 4 shall not liberate gases which combine with the atmosphere to form acid or corrosive alkali, nor shall they liberate toxic fumes when burned.

3.3.6 Protective treatment of materials. Materials used which are utilized within the hermetically sealed case are considered suitably protected against corrosion. Requirements normally specified for fungicidal and corrosion protective treatment, anodizing of aluminum parts, and cadmium plating of steel parts are not applicable for internal parts of hermetically sealed instruments.

3.3.6.1 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of

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climatic and environmental conditions shall be avoided.

3.4 DESIGN AND CONSTRUCTION.

3.4.1 General. The indicator shall be designed to accurately follow up and indicate the Mach number function as computed by and received from the Computer, True Airspeed and Mach Number, Type C-2A. The essentially logarithmic output shall be linearized for presentation on the dial arrangement depicted.

3.4.2 Design. The indicator assembly shall consist of a follow-up synchro, servo motor, and an integral amplifier; designed for operation from the Mach number channel output of the Type C-2A Computer. The servo system shall be so designed as to have a scale position corresponding to each rotor position of the transmitter synchro, regardless of the Mach number at which power is applied.

3.4.3 Hermetical sealing. The indicator shall be hermetically sealed. A tear strip shall be provided to facilitate removal and resealing of the mechanism. There shall also be a filling tube of a malleable variety of metal which shall be capable of being formed into a slot provided on the rear of the case so as to be flush with the rear surface when not in use.

3.4.3.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and containing not more than .006 mg of water vapor per liter (dewpoint -65°C at the filling pressure). The filling medium shall be 100 percent helium or a mixture of 88 to 92 percent nitrogen with the remainder helium. The absolute pressure of the filling medium shall be between one-half and one atmosphere. Wherever practicable the 100 percent helium shall be used.

3.4.4 Construction. The indicator shall be so constructed as to withstand without failure the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

3.4.5 Time delay. The indicator shall be provided with an electro-mechanical device which will delay operation for 45 $\pm$ seconds after power has been applied.

3.5 Performance. The indicator shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperature. Temperatures ranging from -54° to $+71^{\circ}$C.

b. Altitude. Pressures ranging from 30 inches Hg down to 1.68 inches Hg absolute approximately 65,000 feet altitude.

c. Salt fog. Exposure to salt sea atmosphere.

d. Vibration. Vibration incident to service use.

e. Fungus. Fungus growth as encountered in tropical climates.

f. Dust (fine sand). Sand and dust particles as will be encountered in desert areas.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

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3.7 Standard electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400.

3.8 Case. The case shall conform to MS33549 and figure 1 and shall be finished in durable dull black. The case shall be constructed of a suitable material applicable to hermetic sealing, uniform in texture, and which shall provide a smooth external surface.

3.8.1 Cover glass. The cover glass shall be suitable for hermetic sealing, clear, flat, and free from flaws obstructive to normal reading of the indicator dial(s). The cover glass shall be in accordance with MS28105 as pertains to optical quality. The dimensions shall be as required for this application.

3.8.2 Dial to cover glass distance. The distance between the inside surface of the cover glass and the main dial shall not exceed 0.10 inch.

3.9 Instrument dial. The instrument dial shall conform to figure 2.

3.9.1 Visibility of dial markings. All graduations (a minimum of 0.062 of the shortest graduation), numerals, pointers, and other markings on the dial shall be readable from any point within the frustum of a cone, the side of which makes an angle of not less than 30° with a perpendicular to the dial, and the small diameter of which is the aperture of the dial.

3.9.2 Dial finish. The background of the main dial shall be finished in durable dull black. The background of the subdial shall be finished in dull gray, color No. 3635 or TT-C-595.

3.9.2.1 Fluorescent luminescent markings. The following markings shall be finished with fluorescent luminescent material in accordance with MIL-L-25142, type I or III, as applicable.

	Height or length Inches +0.016	Width of line or graduation Inches +0.005
Numerals (Main Dial)	0.250	0.030
.1 Mach Graduations (Main Dial)	0.250	0.030
.05 Mach Graduations (Main Dial)	0.188	0.030
.01 Mach Graduations (Main Dial)	0.156	0.020
Lettering "MACH"	0.156	----
Lettering "HUNDREDTHS"	0.078	----
Settable Mach Index (Main Dial)	Figure 2	----
Stationary "Lubberline" Index	Figure 2	----
Lettering on Setting Knob "Set Indices"	0.094	----

3.9.2.2 Durable dull black. The following markings shall be finished in durable dull black:

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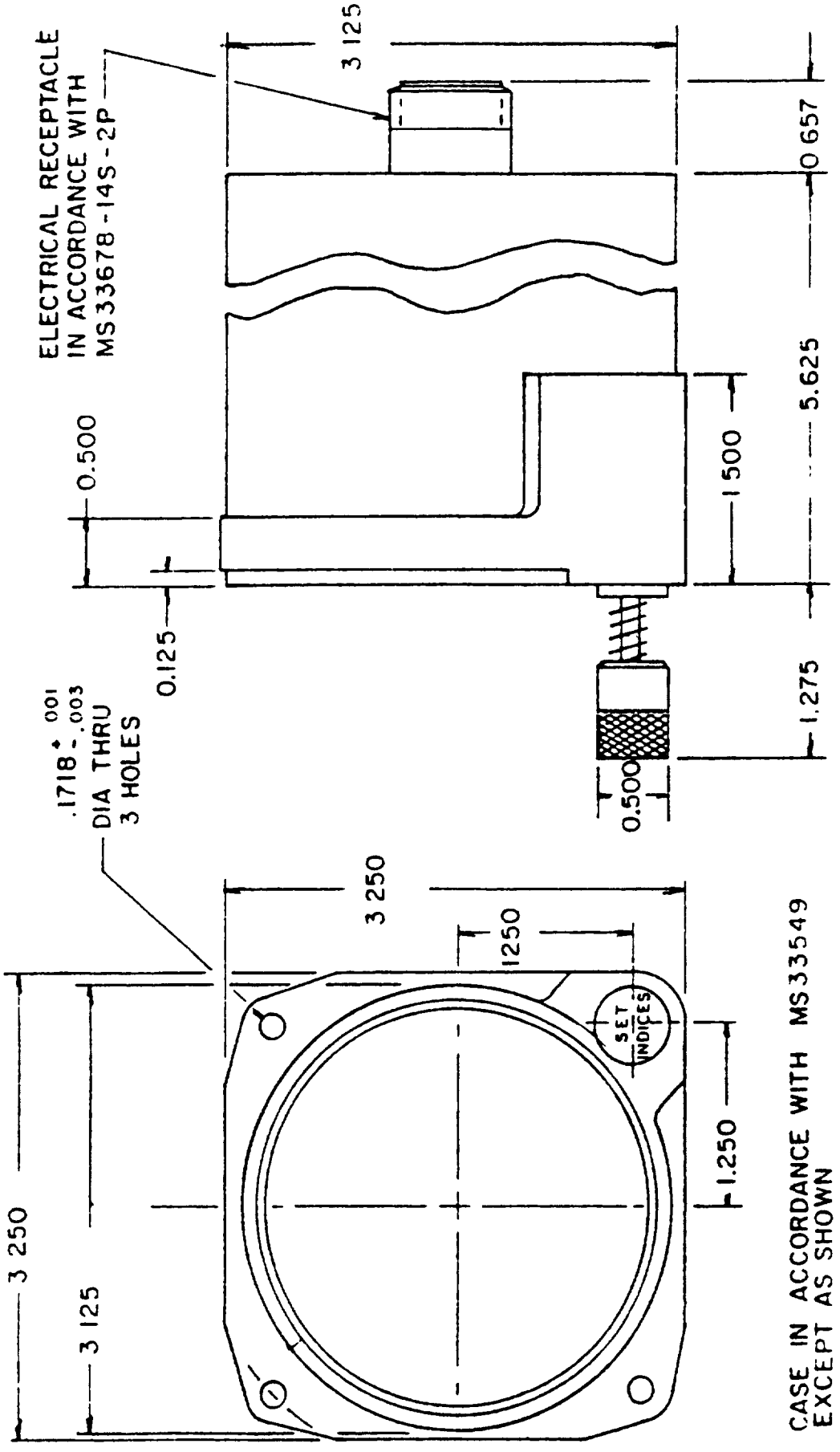


FIGURE 1. Case.

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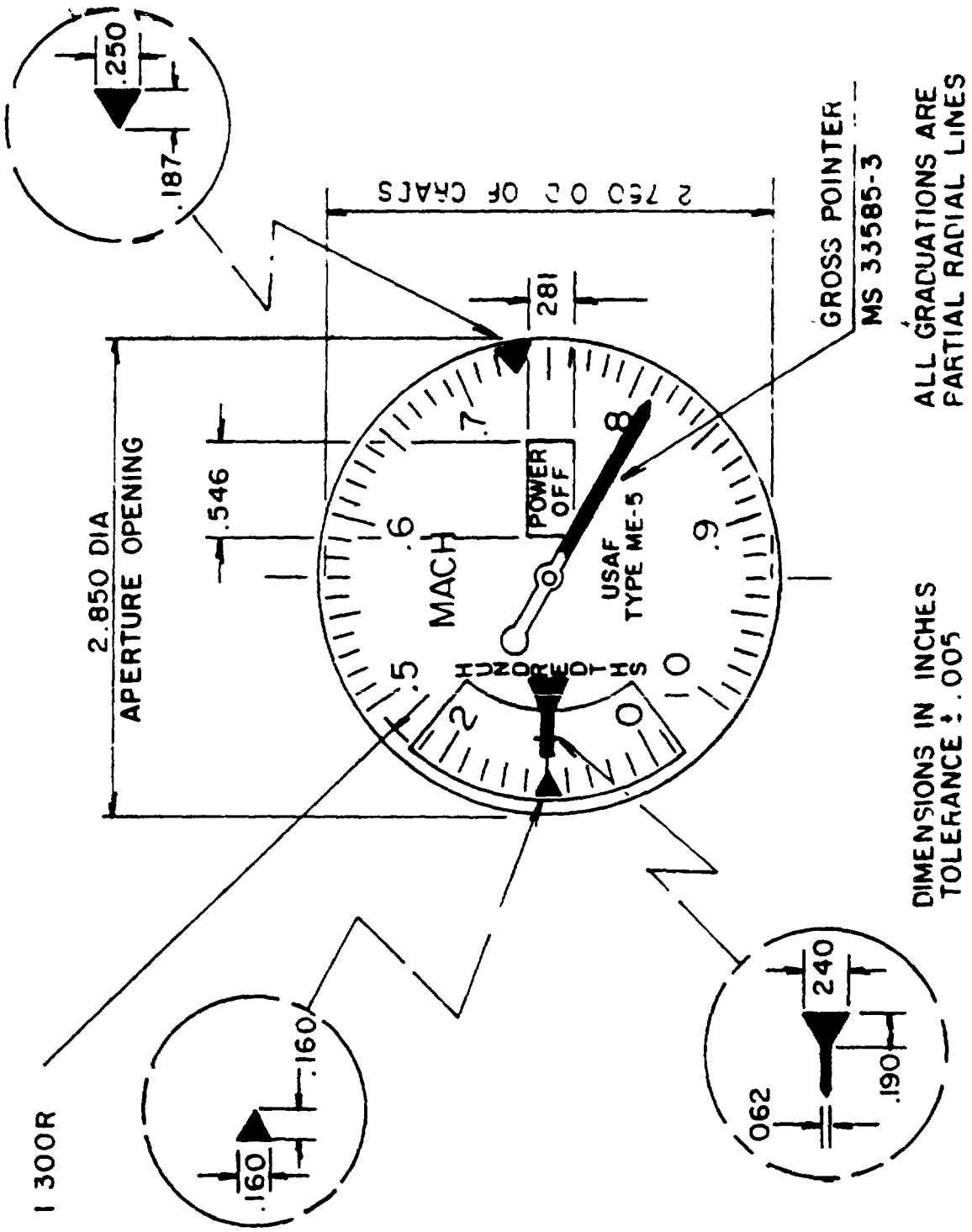


FIGURE 2. Dial

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	Height or Length Inches +0.016	Width of Line of Graduation Inches +0.005
Numerals (Sub-Dial)	0.250	0.030
.01 Mach Graduations (Sub-Dial)	0.219	0.025
.002 Mach Graduations (Sub-Dial)	0.156	0.015
"USAF" and "TYPE ME-5"	0.469	0.010

3.10 Mach number setting indices. The mach number setting indices shall be as shown on figure 2.

3.10.1 Main dial setting index. There shall be a settable index on the Mach number scale, adjustable over the entire scale from 0.5 to 1.0 Mach number. It shall be operable from a knob located in the lower right hand corner of the face of the bezel.

3.10.2 Sub dial setting index. There shall be an index provided on the rotating sub Mach number scale. It shall be settable over the entire range of the sensitive scale and operable by depressing the knob described in 3.10.5. When this index is set at any value of Mach number, it shall rotate and retain its location, with the scale.

3.10.3 Setting. The setting of either index shall not affect the setting of the other index in any way. Operation of the knob and associated indices shall be smooth and free of binding or erratic motion.

3.10.4 Setting sensitivity. The gross setting and the sensitive setting indices shall have a minimum setting of Mach 0.005 and Mach 0.001 respectively. The indices shall turn in the direction of increasing function for clockwise operation of the setting knob.

3.11 Pointers. The gross pointer shall be in accordance with MS33585-3 and the settable indices shall be in accordance with figure 2. Gross pointer length shall be such that the tip shall come to, but not overlap, the shortest graduation.

3.12 Limitation of index movement. Stops shall be provided for the gross setting index preventing movement of the setting index more than $10^{o} + 5^{o}$ beyond the two extremes of the gross scale. There shall be no restraint on the sensitive setting index at any dial position.

3.13 Damping. The indicator shall be essentially critically damped with maximum inertial effects. Response shall be approximately .05 Mach per second.

3.14 Bearings and pivots. Moving elements in the mechanism shall rotate on hardened, ground, and highly polished pivot bearings; or highly polished, first quality, sapphire bearings, or both, or other materials previously approved by the contracting activity.

3.15 Power failure indication. A power failure warning shall be incorporated within the dial aperture of the instrument; design details of the warning indication shall be subject to the approval of the contracting activity. A power failure shall be indicated when primary power to the amplifier is interrupted.

3.16 Radio noise interference. The indicator shall be provided with the necessary noise suppression features to suppress its radio noise within the limits

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specified for conducted and radiated radio noise in MIL-I-6181. In particular, these provisions shall include adequate filters and enclosing case construction which will minimize noise leakage through joints, seams, and mating surfaces. The volume and weight of the filtering components required shall be minimized by the application of proper electrical and mechanical design and construction.

3.17 Electrical connectors. The electrical connectors used on the indicator shall be as shown on figure 1.

3.18 Screw threads. Unless specified otherwise, the threads of all machine screws, 0.060 or larger in diameter, shall conform to MIL-S-7742.

3.19 Lubrication. Lubrication if required shall be accomplished in accordance with MIL-STD-838. Lubricants shall conform to applicable specification unless otherwise approved by the contracting activity.

3.20 Weight. The weight of the indicator shall not exceed 2.75 pounds.

3.21 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.22 Installation.

3.22.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installation of the indicator. They shall be No. 6-32NC-2 round head machine screws, having a durable dull black finish. The length shall be sufficient to permit mounting on a panel 0.188 inch thick.

3.22.2 Envelope. An envelope furnished by the contractor containing mounting screws shall be packed with each indicator, and shall be marked with the following information.

"IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS"

3.23 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner in accordance with MIL-STD-454, requirement 9. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.23.1 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.23.2 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw, bolt or threads.

3.23.3 Gears. Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

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3.23.4 Cleaning. The indicator shall be thoroughly cleaned of loose, spatter, or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.24 Recycled and reclaimed materials. Recycled and reclaimed materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Acceptance tests (see 4.3).

b. First article inspection (see 4.6).

4.3 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests

4.3.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.5, TEST METHODS, of this specification:

a. Examination of product.

b. Dielectric strength.

c. Scale error at room temperature.

d. Friction.

e. Repeatability.

f. Magnetic effect.

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g. Position error.

h. Sealing.

i. Time delay.

j. Low temperature operation.

k. High temperature operation.

4.3.2 Sampling tests. Five indicators selected at random from each 500 or less on the contract or order shall be subjected to the following tests, as desribed under 4.5, TEST METHODS, of this specification:

a. Low temperature exposure.

b. High temperature exposure.

c. Vibration error.

d. Power consumption.

e. Position error.

f. Vibration failure.

4.3.2.1 Rejection and retest. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them.

4.4 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test condition specified in 4.4.1 of this specification.

4.4.1 Standard atmosphere conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified condition.

4.4.2 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.4.3 Attitude. Unless otherwise specified, the indicator shall be tested in normal operating position.

4.4.4 Tolerances. Unless specified otherwise, test points and tolerances shall be in accordance with tables I and II.

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TABLE I. Tolerance

Tests	Tolerances
Friction	0.001
Position Error	0.001
Repeatability	0.001
After Environmental	0.002
Low Temperature Operation	0.002
High Temperature Operation	0.002
Vibration Error	0.001
Vibration Failure	0.001

4.4.5 Test standards. The indicators shall be tested against a transmitting standard with allowances for errors of the standard. The test standard shall consist of a stand with a bracket for mounting of a synchro transmitter in such a manner that the synchro rotor position can be varied with respect to the stator. The device shall have provision for adjusting and indicating angular movement of the synchro rotor with respect to the stator within 0.1^{o} at any reading, and with a rotation of 360°.

4.4.6 Synchro transmitter. The standard test transmitter shall be a Kollsman Type No. 1377-0411, or equivalent, and shall simulate the output of Mach number specified for the Type C-2A computer.

4.4.7 Mach number Output Characteristics of the Type C-2A Computer, True Airspeed and Mach number. The Mach number output synchro is in accordance with MIL-T-5350. The output is provided primarily for operation of a remote type Mach number indicator but the output data provided so that the synchro may be used for other applications if so desired.

TABLE II. Angle Setting.

Mach Number	Electrical Angle Test Setting
.500	0
.520	6.3
.540	12.4
.560	18.3
.580	24.0
.600	29.6
.620	35.0
.640	40.2

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TABLE II. (Cont'd)

Mach Number	Electrical Angle Test Setting
.660	45.3
.680	50.3
.700*	55.2
.720	60.0
.740	64.6
.760	69.2
.780	73.7
.800*	78.1
.820*	82.4
.840*	86.6
.860*	90.8
.880*	94.9
.900*	98.9
.920	102.9
.940	106.8
.960*	110.7
.980	114.5
1.000*	118.2

NOTE: For qualification and sampling tests use all test points listed above. For individual testing use only test points marked with an asterisk. Eight of the test points shall be within ±.001 Mach number, ten of the test points shall be within ±.0014 Mach number, and eleven of the test points shall be within ±.0018 Mach number.

4.4.7.1 Mach Number Synchro rotation. The Mach Number Synchro shall be connected to a source of 26V, 400 cps and to a suitable receptacle as shown in figure 3. The synchro shall be set at electrical zero as a generator at 0.5 Mach number. The stator leads of the synchro are connected to the receptacle so that the phase sequence of pin connections is C.A.D. for increasing Mach number and hence increasing angles as given in Equations (2) and (2A) below.

The rotation of the synchro is expressed by the following equations:

$$\theta = 173.56 \log \frac{\Delta P}{P} + 126.7 \qquad \text{Eq. (2)}$$

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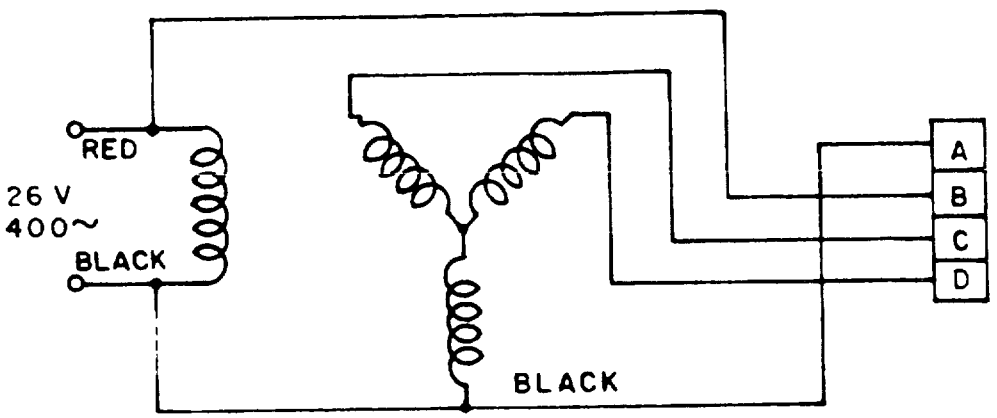


FIGURE 3 Wiring Diagram.

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For Mach numbers less than 0.5, the following equation will yield positive angles:

$$\theta = 173.56 \log \frac{\Delta p}{p} + 486.7 \qquad \text{Eq. (2A)}$$

In equations (2) and (2A), the ration $\frac{\Delta p}{p}$ is related to Mach number as follows:

$$\frac{\Delta p}{p} = (1 + 0.2M^2)3.5 - 1$$

where M = Mach number Eq. (3)

4.4.8 Mach Number vs. Electrical angle. Table II gives the angular position of the Mach number output synchro as a function of Mach number.

4.4.9 Repeater power and pressure indications requirements. Maximum-power required for operation of this repeater shall be 24 va. Mach number readings taken in the course of the testing shall be total pressure readings and errors noted shall be absolute, being the result of both the gross and sensitive readings combined.

4.5 Test Methods.

4.5.1 Examination of product. Each indicator shall be inspected to determine compliance with this specification with respect to materials, workmanship, and markings.

4.5.2 Dielectric strength. A potential of 500V ac at commercial frequency shall be applied between each pin and the indicator case for a minimum period of five (5) seconds. There shall be no insulation breakdown resulting from the application.

4.5.3 Scale error at room temperature. The rotor of the standard synchro transmitter shall be rotated to the desired angular indication. The scale shall then be tested at each of the test points in table II. Errors as read on the Mach scale shall not exceed the tolerances specified in table II.

4.5.4 Friction. The indicator shall be tested for friction error at all the points listed in table II. The rotor of the transmitting synchro shall be set to each test position and held constant while two readings are taken, the first before the indicator is tapped. The difference between any two such readings shall not exceed the tolerance specified in table I.

4.5.5 Repeatability. The instrument shall be capable of repeating the same scale indication or pointer deflection, ignoring the scale error, within the tolerance specified in table I. A minimum of three transmitter setting applications at every other one of the test points in table II shall be made to determine repeatability.

4.5.6 Magnetic effect. The indicator, with power applied, shall be held in various positions with its nearest part 5 inches from the center of a short bar magnet compass whose compensator magnets have been removed, and which shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19° oersted. The various positions shall be so chosen as to denote that position causing the maximum deflection of the compass. Maximum deflection of the magnet shall not exceed 2°. The magnetic effect test may be conducted as a sampling test instead of an individual test.

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4.5.7 Position error. The scale reading taken while the indicator is held in any position, and while it is being tapped, shall not differ from its reading when held in the normal position by more than the tolerance specified in table I. At least one of these positions shall be at approximately 45° with the horizontal, simulating a 45° climb or dive, or both. This test shall be conducted at every other one of the points listed in table II. This test may be combined with the friction error test.

4.5.8 Sealing. The indicator case shall be tested for leaks using a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of one atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 0.1 micron cubic foot per hour.

4.5.9 Low temperature operation. The indicator shall be placed in a chamber and maintained at a temperature of -54° ±2°C for a period of 4 hours with no power applied. While at this temperature, power shall be applied and the indicator tested for scale error as specified in the room temperature scale error test. The errors shall not exceed the tolerance specified in table II by more than the tolerance specified in table I. The repeatability test shall be conducted during this test.

4.5.10 High temperature operation. The indicator shall be placed in a chamber and maintained at a temperature of +71° ±2°C for a period of 4 hours with no electrical power applied. While still at this temperature, power shall be applied to the indicator, and it shall be tested for scale error as specified in the scale error at room temperature tests. The errors at the test points shall not exceed the tolerance in table II by more than the tolerances specified in table I. The repeatability test shall be conducted during this test.

4.5.11 Vibration error. The indicator shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in a horizontal plane, with the plane of the mounting lugs of the indicator in a vertical attitude. The standard transmitter shall be connected to the indicator and set to provide a Mach indication of 0.800. The indicator shall be subjected to a vibration with a circle diameter between 0.003 and 0.005 inch at frequencies from 500 to 3000 cpm and the pointer action noted for resonant points. The maximum total vibration of the pointer shall not exceed the tolerance specified in table I.

4.5.12 Power consumption. With the indicator connected in the test circuit and with power applied, the power consumption shall not exceed 15 va with the servo motor running.

4.5.13 Position error. This test shall be conducted at every other point marked with an asterisk in table II, beginning with the first asterisk. The indicator reading, taken while it is held in any desired position and being tapped, shall not differ from its reading while being held in any other position by more than the amount specified in table I.

4.5.14 Vibration failure. This test shall be conducted in accordance with method 514.2, Equipment Category b.1, procedure I, table 514.2-11 figure 514.2-2 curve C from 5-52 HZ of MIL-STD-810C.

4.5.15 Life test. The indicator shall be subjected to a life test of 320 hours actual operation with 6400 cycles of the signal input. The life test consists of

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20 days of 24 hour duration. The test will begin when electrical power is applied. After 4 hours at rated electrical power, the power shall be turned off and remain off for a period of 2 hours. This procedure of power on for 4 hours and off for 2 hours shall be repeated for the specified 20 days. During all periods when the power is on the instrument shall be cycled full range at least once every 3 minutes. There shall be no damage as a result of this test.

4.5.16 Altitude. The indicator shall be placed within the test chamber and the absolute internal pressure reduced to 1.68 inches Hg (corresponding 65,000 feet above sea level) and an ambient temperature of $-54^{\circ} \pm 2^{\circ}$C for a continuous period of 4 hours. No damage shall result from this test.

4.5.17 Low temperature exposure. This test shall be conducted in accordance with method 502.1 of MIL-STD-810C. After the 24 hour period the indicator shall be tested for scale error at room temperature. Scale errors shall not exceed the tolerances specified for the routine low temperature operation test. There shall be no damage resultant which will adversely affect subsequent operation of the indicator.

4.5.18 High temperature exposure. This test shall be conducted in accordance with method 501.1 of MIL-STD-810C. After the indicator has returned to room temperature, it shall meet the requirements for room temperature scale error.

4.5.19 Radio noise interference. This test shall be conducted in accordance with MIL-I-6181 in the frequency range of 150 kc to 150 mc.

4.5.20 Warm-up time. The indicator shall be connected within the test circuit, and shall have produced the desired reading within 45 seconds following the application of power.

4.5.21 Damping and response. With power on and test connections properly made in accordance with 4.3.5 the transmitter synchro shall be set at 0° (0.5 Mach number). The transmitting synchro shall then be de-energized and the rotor displaced to 98.9° (0.9 Mach number). The transmitting synchro shall then be energized and the elapsed time consumed by the indicator in adjusting its indication from 0.5 Mach number to 0.9 Mach number shall not exceed 7 ± 1 seconds. The maximum amount of overswing shall not exceed 0.002 Mach number. These limiting values shall apply at high and low temperatures. Response of pointer and subdial shall be smooth and without evidence of binding or erratic motion.

4.5.22 Acceleration. The indicator with power applied shall be mounted dynamically similar to service installation on a contrifugal-type test stand and accelerated first in the vertical axis and then in each of the two axes that are perpendicular to the vertical axis and to each other. An acceleration of 8g shall be applied for 10 minutes. No damage to the indicator shall result from this test.

4.5.23 Thermal shock cycling. The indicator shall be alternately subjected to a total of 8 cycles of immersion in tap water maintained at $85^{\circ} \pm 2^{\circ}$C and $5^{\circ} \pm 2^{\circ}$C. The length of time of each immersion shall be 30 minutes. Not more than 5 seconds shall elapse between the immersions. No evidence of moisture penetration or damage shall result from this test. Following this test, the indicator shall be subjected to the test specified in 4.4.8.

4.5.24 Salt fog. This test shall be conducted in accordance with the procedure 509.1 of MIL-STD-810C for a period of 48 hours. At the end of the 50 hour test

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period, there shall be no evidence of external corrosion or discoloration as a result of this test.

4.5.25 Time delay. After the instrument has been in a unenergized condition for more than 5 minutes, power shall be applied. During a period of 45 seconds after power application, the instrument shall not operate.

4.6. First article inspection. The first article inspection samples shall consist of three indicators. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization. The first article inspection shall consist of all test specified under 4.4 TEST METHODS.

4.6.1 First article required. Prior to actual procurement, the product which this specification covers shall, pass the first article inspection specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same first article inspections.

4.7 Inspection of packaging. The inspection of the preservation and interior package marking shall be in accordance with group A and B quality conformance requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified. (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

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5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The position indicator covered by this specification is intended for use in high speed subsonic bomber aircraft, where sensitive Mach number indication is required and where there is a Type C-2A Computer available as a Mach number signal output.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. The quantity, part number and class number of the instrument desired

c. Whether sampling plan B tests are to be conducted

d. Reinspection date marking

e. Level of packaging (5.1) and packing (5.2) desired.

6.3 First article. A first article is required and shall be tested under the appropriate provision of 7-104.55 of the Defense Acquisition Regulation. First article inspection are described in 4.1 and 4.4.

6.4 Definitions.

6.4.1 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal between vitric or metallic materials. A hermetic seal is not

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intended by this definition to include seals effected by means of gaskets.

6.5 Subject term (key word) listing

Indicator
Mach Number
Type ME-5

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Air Force -71

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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